AYA MLOUKI

Junior Software Engineer @ENSIAS

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EDUCATION

ENSIAS – ÉCOLE NATIONALE SUPÉRIEURE D'INFORMATIQUE ET D'ANALYSE DES SYSTÈMES **Rabat, Morocco**
State Engineering Degree in Software Engineering **2024 – Present**

EXPERIENCE

FULL-STACK DEVELOPER | INTERN **JULY 2025 – AUGUST 2025 (2 MONTHS)**
DECTRONIC **Mohammedia, Morocco**

- Developed a web-based industrial project management platform with multi-role workflows, 7 milestones, validations, notifications, and Kanban dashboards
- Designed a REST API with Spring Boot, Spring Security, JWT, RBAC, JPA/Hibernate, DTOs, Bean Validation, migrations, and MySQL to secure and structure data
- Developed the React.js front-end and containerized the application with Docker/Docker Compose to facilitate integration and deployment

Skills: *Spring Boot, Spring Security, JWT, REST API, JPA, Hibernate, React.js, MySQL, Docker, Docker Compose, Bean Validation, RBAC*

FULL-STACK & AI DEVELOPER | INTERN **JULY 2026 – AUGUST 2026 (2 MONTHS)**
DIGITAL BOX TECHNOLOGIES **Rabat, Morocco**

- Developed SentinelAI, a multi-camera physical security platform with real-time detection, a rule engine, LOW-to-CRITICAL incidents, and multi-role management
- Designed Go microservices with JWT, RBAC, GORM/MariaDB, RabbitMQ, and WebSocket for asynchronous communication and real-time notifications
- Developed the Python AI pipeline with YOLOv8, BoTSORT, and IoU-based person-weapon association, then integrated MJPEG streaming into the React.js/TypeScript front-end with Docker Compose

Skills: *Go, Microservices, RabbitMQ, JWT, RBAC, GORM, MariaDB, WebSocket, Python, YOLOv8, BoTSORT, React.js, TypeScript, Docker, Docker Compose*

PROJECTS

FoundersSync – Co-Founder Matching Platform **APRIL 2026 – JUNE 2026**

- Designed a collaborative platform for creating virtual startups, finding co-founders, and collaborating through messaging, news feeds, and real-time notifications
- Developed 6 Spring Boot microservices with Spring Cloud, RabbitMQ, PostgreSQL/MongoDB, and JWT/RBAC, with file storage on AWS S3
- Implemented the Next.js 14/Material UI front-end with Socket.IO and deployed the platform on AWS using Terraform and Docker Compose

Skills: *Java, Spring Boot, Spring Cloud, Node.js, Socket.IO, Next.js, TypeScript, PostgreSQL, MongoDB, RabbitMQ, Docker, Terraform, AWS, JWT*

Business Intelligence Dashboard for Performance Management **APRIL 2026**

- Designed a BI solution combining a data warehouse, ETL, interactive dashboards, and machine learning for decision support
- Developed a constellation-schema data warehouse with SQL Server and an ETL pipeline, then created Power BI dashboards for KPI monitoring
- Integrated scikit-learn models for forecasting, anomaly detection, classification, and segmentation, exposed through Flask

Skills: *SQL Server, Power BI, Python, Flask, scikit-learn*

DermAI – Intelligent Skin Lesion Classification System **JUNE 2026**

- Developed an AI-powered web application for classifying skin lesions from dermoscopic images and clinical data
- Designed a multimodal Deep Learning model combining DenseNet121 for dermoscopic images and an MLP for clinical data, with modality fusion using PyTorch
- Integrated the model with Flask/React and Grad-CAM for prediction explainability and automated PDF report generation

Skills: *Python, PyTorch, Deep Learning, DenseNet121, MLP, Multimodal Learning, Grad-CAM, Flask, React, ISIC 2019*

CERTIFICATIONS

- **AWS Academy Graduate – Cloud Foundations** **DECEMBER 2025**
- **Cambridge English Certificate** **FEBRUARY 2021**

LANGUAGES

- **English** : C1 **French** : C1 **Arabic** : Native